

Task 5: Auto Tagging Support Tickets Using LLM

Objective

Automatically categorize customer support tickets into predefined tags using a Large Language Model (LLM). Compare zero-shot classification against few-shot learning, outputting top-3 predictions per ticket.

Files

File	Description
notebook.ipynb	Complete analysis with zero-shot vs few-shot comparison
src/data_loader.py	Synthetic dataset generation and prompt engineering
src/classifier.py	Zero-shot/few-shot classification utilities
src/visualizations.py	Reusable plotting functions
outputs/figures/	Generated plots (PNG, 200 DPI)

Quick Start

```
pip install -r ../requirements.txt
jupyter notebook notebook.ipynb
```

Note: The first run will download `facebook/bart-large-mnli` (~1.6 GB). This may take 5-10 minutes depending on your internet speed.

Key Results

- **Zero-Shot Accuracy:** ~60-70% (no training data required)
- **Few-Shot Accuracy:** ~70-80% (with just 2 examples per class)
- **Best Performing Tags:** Billing, Technical
- **Most Challenging:** Feature Request, General Inquiry
- **Top-3 Predictions:** Enable human-in-the-loop workflows even when top-1 is wrong

Tech Stack

- Python 3.10+
- pandas, numpy
- matplotlib, seaborn
- transformers (Hugging Face)
- scikit-learn (metrics)
- Jupyter Notebook